

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): MW843UST, MX842UST

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0002-b002	1.4.2

Version History

Module Version	Date	Notes
1_0_1_0	2/16/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='MW843UST')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='MW843UST')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4352, Model='MW843UST')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ¹	Value '4:3'	Value '16:9'	Value '16:10'
	'Auto'	'Real'	
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AudioMute ¹	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage ¹	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command AVMute ^{2,3}	Value 'On'	Value 'Off'	
Qualifier Key 'Type'	Qualifier Value 'Audio'	Qualifier Value 'Video'	Qualifier Value 'Audio/Video'
# AVMute example InterfaceName.Set('AVMute', 'On', {'Type': 'Audio'})			
Command Freeze ¹	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer 1' ¹ 'Video' ¹ 'VGA 1' ³ 'Card Reader' ³	Value 'Computer 2' ³ 'HDMI' 'VGA 2' ³ 'LAN Display' ³	Value 'S-Video' 'Network' ¹ 'CVBS' ³ 'USB Display' ³
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command LampMode ¹	Value 'Normal'	Value 'Economic'	Value 'SmartEco'
# LampMode example InterfaceName.Set('LampMode', 'Normal')			

Command MenuNavigation ¹	Value 'Menu On' 'Down' 'Enter'	Value 'Menu Off' 'Right'	Value 'Up' 'Left'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu On')			
Command PictureMode ¹	Value 'Presentation' 'Cinema'	Value 'sRGB' 'User 1'	Value 'Dynamic' 'User 2'
# PictureMode example InterfaceName.Set('PictureMode', 'Presentation')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute ¹	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume ¹	Value 'Up'	Value 'Down'	
# Volume example InterfaceName.Set('Volume', 'Up')			

¹ RS-232 support only.² This command requires the Type parameter.³ Ethernet support only.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio ¹	Value '4:3' 'Auto'	Value '16:9' 'Real'	Value '16:10'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute ¹	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command AVMute ^{2,3}	Value 'On'	Value 'Off'	
Qualifier Key 'Type'	Qualifier Value 'Audio'	Qualifier Value 'Video'	Qualifier Value 'Audio/Video'
# AVMute examples InterfaceName.Update('AVMute', {'Type': 'Audio'}) Value = InterfaceName.ReadStatus('AVMute', {'Type': 'Audio'}) InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command ErrorStatus ²	Value 'Fan Warning'	Value 'Fan Error'	Value 'Lamp Warning'

Global Scripter Module
Communication Sheet

	'Lamp Error'	'Temp Warning'	'Temp Error'
	'Cover Warning'	'Cover Error'	'Filter Warning'
	'Filter Error'	'Other Warning'	'Other Error'
	'Normal'	'Multiple Errors/Warnings'	
# ErrorStatus examples InterfaceName.Update('ErrorStatus') Value = InterfaceName.ReadStatus('ErrorStatus') InterfaceName.SubscribeStatus('ErrorStatus', None, FeedbackHandler)			
Command Freeze ¹	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'Computer 1' ¹ 'Video' ¹ 'VGA 1' ³ 'Card Reader' ³	Value 'Computer 2' ³ 'HDMI' 'VGA 2' ³ 'LAN Display' ³	Value 'S-Video' 'Network' ¹ 'CVBS' ³ 'USB Display' ³
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampHours	Value 0 – 99999		
# LampHours examples InterfaceName.Update('LampHours') Value = InterfaceName.ReadStatus('LampHours') InterfaceName.SubscribeStatus('LampHours', None, FeedbackHandler)			
Command LampMode ¹	Value 'Normal'	Value 'Economic'	Value 'SmartEco'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command PictureMode ¹	Value 'Presentation' 'Cinema'	Value 'sRGB' 'User 1'	Value 'Dynamic' 'User 2'
# PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	Value 'Cooling Down' ³

		'Warming Up' ³
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)		
Command	Value	Value
VideoMute ¹	'On'	'Off'
# VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)		
Command	Value	
VolumeStatus ¹	0 – 10	
# VolumeStatus examples InterfaceName.Update('VolumeStatus') Value = InterfaceName.ReadStatus('VolumeStatus') InterfaceName.SubscribeStatus('VolumeStatus', None, FeedbackHandler)		

¹ RS-232 support only.² This command requires the Type parameter.³ Ethernet support only.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

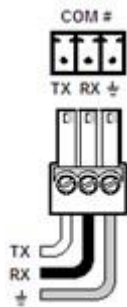
Data Bits: 8

Parity: None

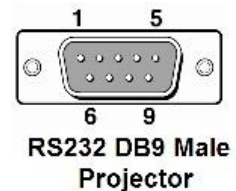
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	4352
Logon Credentials Supported:	Yes
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands (Serial)

Aspect Ratio 16:10	\x0D*asp=16:10#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Real	\x0D*asp=REAL#\x0D
Audio Mute Off	\x0D*mute=off#\x0D
Audio Mute On	\x0D*mute=on#\x0D
Auto Image None	\x0D*auto#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input HDMI	\x0D*sour=hdm#\x0D
Input S-Video	\x0D*sour=svid#\x0D
Input Computer 1	\x0D*sour=RGB#\x0D
Input Computer 2	\x0D*sour=RGB2#\x0D
Input Network	\x0D*sour=network#\x0D
Input Video	\x0D*sour=vid#\x0D
Lamp Mode Economic	\x0D*lampm=eco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Lamp Mode SmartEco	\x0D*lampm=seco#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Menu Off	\x0D*menu=off#\x0D
Menu Navigation Menu On	\x0D*menu=on#\x0D
Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D
Picture Mode Cinema	\x0D*appmod=cine#\x0D
Picture Mode Dynamic	\x0D*appmod=dynamic#\x0D
Picture Mode Presentation	\x0D*appmod=preset#\x0D
Picture Mode User 1	\x0D*appmod=user1#\x0D
Picture Mode User 2	\x0D*appmod=user2#\x0D
Picture Mode sRGB	\x0D*appmod=srgb#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D

Global Scripter Module Communication Sheet

Revision: 2/16/2018

Volume Down	\x0D*vol=-#\x0D
Volume Up	\x0D*vol=+\#\x0D

Appendix B. Update Commands (Serial)

Aspect Ratio	\x0D*asp=?#\x0D
Audio Mute	\x0D*mute=?#\x0D
Freeze	\x0D*freeze=?#\x0D
Input	\x0D*sour=?#\x0D
Lamp Hours	\x0D*ltim=?#\x0D
Lamp Mode	\x0D*lampm=?#\x0D
Picture Mode	\x0D*appmod=?#\x0D
Power	\x0D*pow=?#\x0D
Video Mute	\x0D*blank=?#\x0D
Volume Status	\x0D*vol=?#\x0D

Appendix C. Set Commands (Ethernet)

AV Mute Off Type Audio	%1AVMT 20\x0D
AV Mute On Type Audio	%1AVMT 21\x0D
AV Mute Off Type Audio/Video	%1AVMT 30\x0D
AV Mute On Type Audio/Video	%1AVMT 31\x0D
AV Mute Off Type Video	%1AVMT 10\x0D
AV Mute On Type Video	%1AVMT 11\x0D
Input CVBS	%1INPT 22\x0D
Input Card Reader	%1INPT 51\x0D
Input HDMI	%1INPT 31\x0D
Input LAN Display	%1INPT 52\x0D
Input S-Video	%1INPT 21\x0D
Input USB Display	%1INPT 53\x0D
Input VGA 1	%1INPT 11\x0D
Input VGA 2	%1INPT 12\x0D
Power Off	%1POWR 0\x0D
Power On	%1POWR 1\x0D

Appendix D. Update Commands (Ethernet)

AV Mute Type Audio	%1AVMT ?\x0D
AV Mute Type Audio/Video	%1AVMT ?\x0D
AV Mute Type Video	%1AVMT ?\x0D
Error Status	%1ERST ?\x0D
Input	%1INPT ?\x0D
Lamp Hours	%1LAMP ?\x0D
Power	%1POWR ?\x0D